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Task: Pseudocode for IDA* Search Algorithm

Pseudocode for IDA* Search Algorithm

ALGORITHM IDA*(start, goal)

1. threshold ← h(start)

2. LOOP:

    result ← DFS_LIMITED(start, 0, threshold, goal)

    IF result = FOUND:

        RETURN solution path

    IF result = INFINITY:

        RETURN FAILURE

    threshold ← result

FUNCTION DFS_LIMITED(node, g, threshold, goal)

1. f ← g + h(node)

2. IF f > threshold:
    RETURN f

3. IF node = goal:
    RETURN FOUND

4. minimum ← INFINITY

5. FOR each successor of node:
    new_g ← g + cost(node, successor)

    result ← DFS_LIMITED(
        successor,
new_g,
threshold,
goal)

    IF result = FOUND:
        RETURN FOUND

    IF result < minimum:
minimum ← result

6. RETURN minimum